



Lecture 22.04: Operations on Functions: Scalar Multiplication & Addition

 Python 3.10+

 Numpy Numerical Computing

 Status Completed

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1. Algebraic Operations on Functions

Given two real-valued functions $f, g : \mathbb{R} \rightarrow \mathbb{R}$ and a scalar constant $c \in \mathbb{R}$:

1. Function Addition:

$$(f + g)(x) = f(x) + g(x)$$

2. Scalar Multiplication:

$$(c \cdot f)(x) = c \cdot f(x)$$

3. Linear Combination:

$$h(x) = c_1 f(x) + c_2 g(x)$$

The domain of the sum $(f + g)$ is the intersection of their domains:

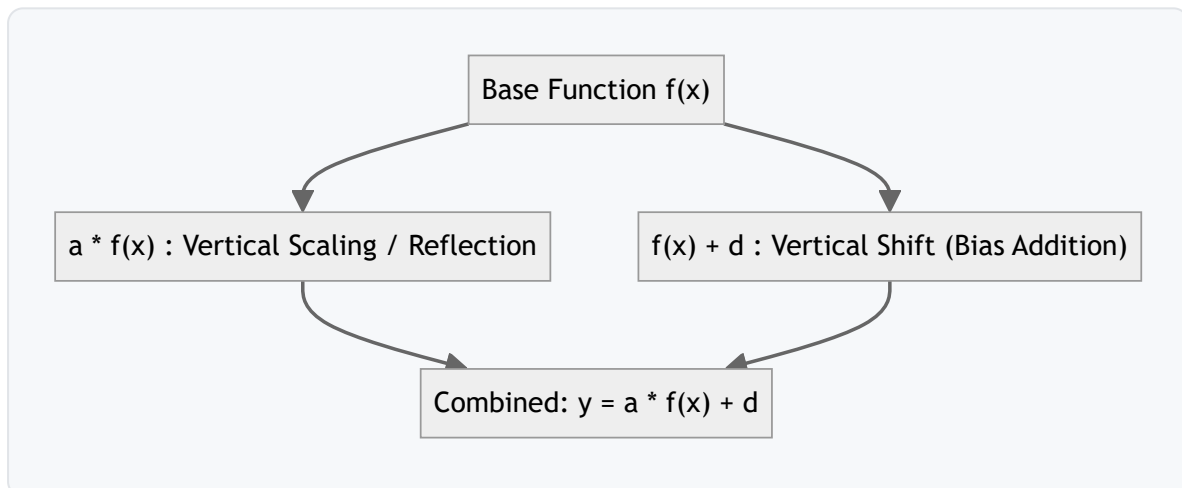
$$\text{Dom}(f + g) = \text{Dom}(f) \cap \text{Dom}(g)$$



2. Geometric Effects: Vertical Transformations

When scalar operations are applied **outside** the function evaluation, they manipulate the graph vertically:

Transformation	Equation	Geometric Effect
Vertical Stretch	$y = a \cdot f(x)$ with $a > 1$	Graph stretched away from the x-axis by factor a .
Vertical Compression	$y = a \cdot f(x)$ with $0 < a < 1$	Graph compressed toward the x-axis by factor a .
Vertical Reflection	$y = -f(x)$	Graph reflected across the x-axis.
Vertical Shift Up	$y = f(x) + d$ with $d > 0$	Graph shifted upward by d units.
Vertical Shift Down	$y = f(x) - d$ with $d > 0$	Graph shifted downward by d units.



3. Machine Learning Application: Weighted Neurons & Residual Connections

1. **Perceptron Output:** A single artificial neuron computes a linear combination of feature functions plus a vertical bias:

$$y = \sum_{i=1}^m w_i x_i + b$$

2. **Residual Networks (ResNets):** Residual skip connections perform function addition:

$$\mathbf{h}_{l+1} = \mathbf{h}_l + \mathcal{F}(\mathbf{h}_l, W_l)$$

Adding the identity function $\mathbf{h}_l$ prevents vanishing gradients and allows deep networks to train effectively.

4. Python Implementation: Visualizing Vertical Transformations

```
import numpy as np
import matplotlib.pyplot as plt

x = np.linspace(-3, 3, 300)
base = x**2

plt.figure(figsize=(9, 5))
plt.plot(x, base, 'k-', lw=2, label='Base: f(x) = x^2')
plt.plot(x, 2 * base, 'r--', lw=2, label='Vertical Stretch: 2 * f(x)')
plt.plot(x, 0.5 * base, 'g:', lw=2, label='Vertical Compression: 0.5 * f(x)')
plt.plot(x, base + 2, 'b-.', lw=2, label='Vertical Shift Up: f(x) + 2')
plt.plot(x, -base, 'm--', lw=2, label='Vertical Reflection: -f(x)')

plt.axhline(0, color='gray', lw=1)
plt.axvline(0, color='gray', lw=1)
plt.title('Vertical Transformations of Functions', fontsize=12, fontweight='bold')
plt.xlabel('x')
plt.ylabel('y')
plt.grid(True, linestyle=':', alpha=0.6)
plt.legend()
plt.ylim(-5, 10)
plt.show()
```